

## Spazi di funzioni integrabili, spazi $L^p$

$(X, \mathcal{A}, \mu)$  sp. di misura,  $\mathcal{M}(X) = \{f: X \rightarrow [-\infty, +\infty] : f \text{ misurabile}, f(x) \in \mathbb{R} \text{ q.o. } x \in X\}$   
 $\uparrow$   
 $\sigma$ -algebra

Per  $1 \leq p < +\infty$ , sia  $\|\cdot\|_p: \mathcal{M}(X) \rightarrow [0, +\infty)$ ,  $\|f\|_p = \left( \int_X |f(x)|^p d\mu \right)^{1/p} = \|f\|_{L^p(X, \mathcal{A}, \mu)}$   
e definiamo  $L^p(X) = \{f \in \mathcal{M}(X) : \|f\|_p < \infty\}$ .

oss Per  $p=2$ , la norma  $\|\cdot\|_2$  proviene da un prodotto scalare:  $f, g \in L^2(X)$ ,  
 $(f, g) = \int_X f(x) \cdot g(x) d\mu(x)$

Per  $p = +\infty$ , sia  $\|f\|_\infty = \inf \{M \in \mathbb{R}^+ : \mu(\{x \in X : |f(x)| > M\}) = 0\}$ , e definiamo poi  
 $L^\infty(X) = \{f \in \mathcal{M}(X) : \|f\|_\infty < \infty\}$ .

def Diciamo che  $1 \leq p, q \leq +\infty$  sono **HÖLDER CONIUGATI** se  $\frac{1}{p} + \frac{1}{q} = 1 \iff q = \frac{p}{p-1} = p'$

prop (**disuguaglianza di Hölder**). Siano  $p, q$  coniugati e  $f \in L^p(X)$ ,  $g \in L^q(X)$ ,  
allora  $f \cdot g \in L^1(X)$  e inoltre

$$\int_X |f(x) \cdot g(x)| d\mu(x) \leq \|f\|_p \|g\|_q$$

Inoltre per  $\infty > p > 1$  e  $\|f\|_p \neq 0$ , si vale l'uguaglianza  $\exists d > 0$  tale  
 $|g(x)| = d |f(x)|^{p-1} \quad \mu\text{-q.o. } x \in X$ .

dim Ricordo la disuguaglianza di Young:  $\forall x, y > 0$  si ha

$$xy \leq \frac{1}{p} x^p + \frac{1}{q} y^q. \text{ Usando Young:}$$

$$\begin{aligned} \int_X \frac{|f(x)|}{\|f\|_p} \cdot \frac{|g(x)|}{\|g\|_q} d\mu &\leq \int_X \left( \frac{1}{p} \frac{|f(x)|^p}{\|f\|_p^p} + \frac{1}{q} \frac{|g(x)|^q}{\|g\|_q^q} \right) d\mu \\ &= \frac{1}{p} \frac{\int_X |f(x)|^p d\mu}{\|f\|_p^p} + \frac{1}{q} \frac{\int_X |g(x)|^q d\mu}{\|g\|_q^q} \\ &= \frac{1}{p} + \frac{1}{q} = 1 \end{aligned}$$

$$\text{Ritorno a Young: } xy \leq \frac{1}{p} x^p + \frac{1}{q} y^q \iff \frac{x}{y^{q-1}} \leq \frac{1}{p} \frac{x^p}{y^q} + \frac{1}{q}$$

$$\begin{aligned} q-1 = \frac{p}{p-1} - 1 = \frac{1}{p-1} &\iff \frac{x}{y^{1/(p-1)}} \leq \frac{1}{p} \left( \frac{x}{y^{1/(p-1)}} \right)^p + \frac{1}{q} \\ \iff \frac{x}{y^{1/(p-1)}} &\leq \frac{1}{p} t^p + \frac{1}{q} \end{aligned}$$

L'uguaglianza vale  $\iff 1 = t = \frac{x}{y^{1/(p-1)}} \iff y = x^{p-1}$ , dunque, tornando  
ad Hölder vale =

$$\iff \frac{|g(x)|}{\|g\|_q} = \frac{|f(x)|^{p-1}}{\|f\|_p^{p-1}} \quad \mu\text{-q.o. } x \in X$$

$$\iff |g(x)| = \left( \frac{\|g\|_q}{\|f\|_p^{p-1}} \right) |f(x)|^{p-1}, \quad \mu\text{-q.o. } x \in X$$

Caso  $p=1$  e  $q=+\infty$

$$\int_X |f(x)| |g(x)| d\mu \leq \|g\|_\infty \cdot \int_X |f(x)| d\mu = \|f\|_1 \cdot \|g\|_\infty.$$

$\wedge$   
 $\|g\|_\infty$  per def. di  $\|\cdot\|_\infty$  per  $\mu$ -q.o.  $x \in X$

□

prop (disuguaglianza di Minkowski). Sia  $1 \leq p \leq +\infty$  e  $f, g \in \mathcal{L}^p(X)$ , allora  $f+g \in \mathcal{L}^p(X)$  e  $\|f+g\|_p \leq \|f\|_p + \|g\|_p$ .

dim Sia  $1 < p < +\infty$ . Allora  $\|f+g\|_p^p = \int_X |f+g|^p(x) d\mu$

Osservo che  $\forall p \geq 1 \exists C_p > 0$  t.c.  $\forall x, y \geq 0 (x+y)^p \leq C_p (x^p + y^p)$ ,  
 da cui segue che  $f+g \in \mathcal{L}^p(X)$ .

$$\begin{aligned} \|f+g\|_p^p &= \int_X |f(x)+g(x)|^{p-1} \cdot |f(x)+g(x)| d\mu \\ &\leq \int_X |f(x)+g(x)|^{p-1} (|f(x)| + |g(x)|) d\mu \\ &= \int_X |f(x)+g(x)|^{p-1} |f(x)| d\mu + \int_X |f(x)+g(x)|^{p-1} |g(x)| d\mu \\ &\stackrel{\text{Hölder}}{=} \left( \int_X |f(x)+g(x)|^{p-1 \cdot q} d\mu \right)^{1/q} (\|f\|_p + \|g\|_p) \\ &= \left( \int_X |f+g|^p d\mu \right)^{\frac{p-1}{p}} (\|f\|_p + \|g\|_p) \end{aligned}$$

$$\Rightarrow \|f+g\|_p \leq \|f\|_p + \|g\|_p.$$

□

esempio  $X=[0,1]$ ,  $\mu$  misura di Lebesgue,  $\alpha > 0$  parametro,  $1 \leq p < +\infty$ ,  $\mathbb{Q} \cap [0,1] = \{q_n\}_{n \in \mathbb{N}}$

Sia  $f: [0,1] \rightarrow [0,+\infty)$ ,  $f(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{1}{|x-q_n|^\alpha}$

Usando Minkowski nel caso numerabile: posso fare una  $\Sigma$  infinita per c.m.

$$\begin{aligned} \left( \int_{[0,1]} |f(x)|^p dx \right)^{1/p} &\leq \left( \sum_{i=1}^{\infty} \left( \int_{[0,1]} \left| \frac{1}{2^n} \frac{1}{|x-q_n|^\alpha} \right|^p dx \right)^{1/p} \right)^{1/p} \\ &= \sum_{i=1}^{\infty} \left( \int_{-q_n}^{1-q_n} \frac{1}{2^{np}} \frac{1}{|t|^{p\alpha}} dt \right)^{1/p} \\ &\leq \sum_{i=1}^{\infty} \left( \frac{1}{2^{np}} \int_0^1 \frac{1}{|t|^{p\alpha}} dt \right)^{1/p} \\ &= \sum_{i=1}^{\infty} \frac{1}{2^n} \left( 2 \int_0^1 \frac{1}{t^{p\alpha}} dt \right)^{1/p} \\ &\stackrel{\text{devo richiedere}}{=} \sum_{i=1}^{\infty} \frac{1}{2^n} 2^{1/p} \left( \left[ \frac{t^{1-p\alpha}}{1-p\alpha} \right]_0^1 \right)^{1/p} \\ &\stackrel{1-p\alpha > 0}{\Leftrightarrow \alpha < \frac{1}{p}} = \sum_{i=1}^{\infty} \frac{1}{2^n} \cdot \frac{1}{(1-p\alpha)^{1/p}} = \left( \frac{2}{1-p\alpha} \right)^{1/p} < \infty \end{aligned}$$

In conclusione, se  $0 < \alpha < \frac{1}{p}$ ,  $f \in \mathcal{L}^p([0,1])$  e la serie converge q.o.

## Inclusioni tra spazi $\mathcal{L}^p$

1)  $\mu(X) < \infty$ , allora  $1 \leq p < q \Rightarrow \mathcal{L}^q(X) \subseteq \mathcal{L}^p(X)$

dim sia  $f \in \mathcal{L}^q(X)$ :  $\int_X |f|^p \cdot 1 d\mu \stackrel{\text{Hölder}}{\leq} \left( \int_X |f|^q d\mu \right)^{\frac{p}{q}} \left( \int_X 1^{\left(\frac{q}{p}\right)'} d\mu \right)^{\frac{1}{\left(\frac{q}{p}\right)'}}$   
 $\stackrel{\mathcal{L}^{\frac{q}{p}}}{=} \|f\|_q^p \cdot \mu(X)^{\frac{1}{\left(\frac{q}{p}\right)'}} < \infty. \quad \square$

2) se  $f \in \mathcal{L}^p(X) \cap \mathcal{L}^q(X)$ ,  $1 \leq p < q < +\infty$ , allora  $f \in \mathcal{L}^r(X) \quad \forall r \in [p, q]$

dim  $\int_X |f|^r d\mu = \int_X |f|^{tq + (1-t)p} d\mu \stackrel{\text{Hölder}}{\leq} \left( \int_X |f|^q d\mu \right)^t \left( \int_X |f|^p d\mu \right)^{1-t} < \infty$   
 $\stackrel{\mathcal{L}^{q/t}}{\in} \mathcal{L}^{q/t} \cdot \mathcal{L}^{p/(1-t)} \stackrel{\mathcal{L}^{1/(1-t)}}{\in} \mathcal{L}^{1/(1-t)}$

3) se  $f \in \mathcal{L}^p(X) \quad \forall p \geq 1$  (definitivamente) e  $\|f\|_p \leq M < \infty \quad \forall p \geq 1 \Rightarrow f \in \mathcal{L}^\infty(X)$  e  $\|f\|_\infty \leq M$

dim sia  $\varepsilon > 0$ , considero  $\mu(\{x \in X : |f(x)| \geq M + \varepsilon\}) = \int_{\{ \dots \}} 1 d\mu$   
 $\leq \int_{\{ \dots \}} \frac{|f(x)|^p}{(M + \varepsilon)^p} d\mu$   
 $\leq \int_X \frac{|f(x)|^p}{(M + \varepsilon)^p} d\mu$   
 $= \frac{\|f\|_p^p}{(M + \varepsilon)^p}$   
 $\leq \frac{M^p}{(M + \varepsilon)^p} = \left( \frac{M}{M + \varepsilon} \right)^p \xrightarrow{p \rightarrow \infty} 0$   
 $\Rightarrow \|f\|_\infty \leq M. \quad \square$

esempio  $X = (0, 1)$ ,  $\mu = \mathcal{L}^1$ ,  $f: (0, 1) \rightarrow \mathbb{R}$ ,  $f(x) = |\ln(x)|$

Osservo che per  $p \geq 1 \quad \exists c_p > 0$ , finito, tale  $\sqrt{x} |\ln(x)|^p \leq c_p$

Allora  $\int_{(0,1)} |\ln(x)|^p dx \leq \int_{(0,1)} \frac{c_p}{\sqrt{x}} dx < \infty \Rightarrow f \in \mathcal{L}^p(0,1) \quad \forall p \geq 1$ , ma  $f \notin \mathcal{L}^1(0,1)$

## Completezza

oss 1  $\mathcal{L}^p(X)$  è sp. vett. per Minkowski.

oss 2  $\|\cdot\|_p: \mathcal{L}^p(X) \rightarrow [0, +\infty)$  verifica:

i)  $\|\cdot\|_p \geq 0$ ,  $\|f\|_p = 0 \Rightarrow f(x) = 0 \quad \mu\text{-q.o. } x \in X$

ii)  $\|df\|_p = |d| \|f\|_p$

iii)  $\|f+g\|_p \leq \|f\|_p + \|g\|_p$  (Minkowski)

} parliamo di  
SEMINORMA

oss 3 Se  $L^p(X)$  introduciamo la relazione d'eq.  $f \sim g \iff f(x) = g(x) \quad \mu\text{-q.o. } x \in X$ .

Sia  $\mathcal{L}^p(X) = L^p(X) / \sim$ . Allora  $(\mathcal{L}^p(X), \|\cdot\|_p)$  è **SPAZIO NORMATO**.

**TEOREMA**  $\forall p \in [1, +\infty]$ ,  $(\mathcal{L}^p(X), \|\cdot\|_p)$  è uno spazio di Banach (spazio normato completo).

dim. Sia  $f_n \in \mathcal{L}^p(X)$ ,  $n \in \mathbb{N}$ , una succ. di Cauchy, cioè  $\forall \varepsilon > 0 \exists \bar{n} : \forall n, m \geq \bar{n}$   
 $\|f_n - f_m\|_p < \varepsilon$ . Obiettivo: provare che  $\exists f \in \mathcal{L}^p(X)$  t.c.  $\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0$ .

• Osservo che  $\exists k \mapsto n_k \in \mathbb{N}$  t.c.  $\|f_{n_{k+1}} - f_{n_k}\|_p < \frac{1}{2^k}$ .

Considero la funzione  $g(x) = \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)| \in [0, +\infty]$ .

Per Minkowski,  $\|g\|_p \leq \sum_{k=1}^{+\infty} \|f_{n_{k+1}} - f_{n_k}\|_p \leq \sum_{k=1}^{+\infty} \frac{1}{2^k} = 1 \implies g(x) < \infty \quad \mu\text{-q.o. } x \in X$

e  $\sum_{k=1}^{\infty} f_{n_{k+1}}(x) - f_{n_k}(x)$  converge assolutamente  $\mu\text{-q.o. } x \in X$

• Definisco  $f(x) := f_{n_1}(x) + \sum_{k=1}^{\infty} (f_{n_{k+1}}(x) - f_{n_k}(x)) \in \mathcal{L}^p(X)$

$$= \lim_{k \rightarrow \infty} f_{n_k}(x) \quad \text{per q.o. } x \in X$$

• Affermo:  $\lim_{n \rightarrow +\infty} \|f_n - f\|_p = 0$ . Sia  $\varepsilon > 0$ .

$$\int_X |f_n(x) - f(x)|^p d\mu = \int_X \lim_{k \rightarrow +\infty} |f_n(x) - f_{n_k}(x)|^p d\mu$$

$$\stackrel{\text{Fatou}}{\leq} \liminf_{k \rightarrow +\infty} \int_X |f_n(x) - f_{n_k}(x)|^p d\mu \leq \varepsilon \quad \forall n \geq \bar{n}$$

$\exists \bar{n} \in \mathbb{N}$

per la condizione di Cauchy.

• Per il caso  $p = \infty$ :  $f_n, n \in \mathbb{N}$ , succ. di Cauchy in  $\mathcal{L}^\infty(X)$ . Siano  $\forall m, n, k \in \mathbb{N}$

$$A_{mn} = \{x \in X : |f_m(x) - f_n(x)| > \|f_m - f_n\|_\infty\}, \quad \mu(A_{mn}) = 0$$

$$B_k = \{x \in X : |f_k(x)| > \|f_k\|_\infty\}, \quad \mu(B_k) = 0$$

allora

$$E := \bigcup_k B_k \cup \bigcup_{m,n} A_{mn}, \quad \mu(E) = 0$$

Sul complementare  $X \setminus E$ ,  $(f_n)_{n \in \mathbb{N}}$  è unif. di Cauchy e conv. unif.

ad una funzione limitata.  $\square$

corollario Siano  $f_n, f \in L^p(X)$ ,  $n \in \mathbb{N}$ ,  $1 \leq p < +\infty$ , t.c.  $\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0$ . Allora  $\exists$  sottosucc.  $(f_{n_k})_{k \in \mathbb{N}}$  t.c.  $\lim_{k \rightarrow +\infty} f_{n_k}(x) = f(x)$  per  $\mu\text{-q.o. } x \in X$ .

dim  $\lim_{k \rightarrow +\infty} \|f_n - f\|_p = 0 \implies f_n$  di Cauchy

L'idea è di costruire una sottosucc. unif. convergente, come sopra.



## Convergenza forte e debole

$(X, \mathcal{A}, \mu)$  sp. di misura,  $1 \leq p \leq +\infty$ ,  $f_n, f \in L^p(X)$ .

def (convergenza forte). Diciamo che  $f_n \xrightarrow[n \rightarrow \infty]{L^p\text{-forte}} f$  se  $\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0$ .

esempio  $X = [0, 1]$ ,  $p = 1$ , sia  $f_n \in L^1([0, 1])$ ,  $n \in \mathbb{N}$ , come segue:  $f_1 = \chi_{[0, 1]}$ ,  $f_2 = \chi_{[0, 1/2]}$ ,  $f_3 = \chi_{[1/2, 1]}$ ,  $f_4 = \chi_{[0, 1/4]}$ ,  $f_5 = \chi_{[1/4, 1/2]}$ ,  $f_6 = \chi_{[1/2, 3/4]}$ ,  $f_7 = \chi_{[3/4, 1]}$ ,  
 $(f_{2^n+k} = \chi_{[\frac{k}{2^n}, \frac{k+1}{2^n}])$ ,  $k \in \{0, \dots, 2^n-1\}$ .

$$\limsup_{n \rightarrow \infty} f_n(x) = 1, \quad \liminf_{n \rightarrow \infty} f_n(x) = 0 \quad \forall x \in [0, 1]$$

Tuttavia,  $\lim_{n \rightarrow \infty} \|f_n - 0\|_1 = \lim_{n \rightarrow \infty} \int_{[0, 1]} |f_n - 0| d\mu = 0$ , cioè  $f_n \xrightarrow[n \rightarrow \infty]{L^1\text{-forte}} 0$   
(ed  $\exists$  sottosucc. conv. nel esempio  $f_{2^n}$ )

Siano ora  $1 \leq p < +\infty$ ,  $1 < q \leq +\infty$ ,  $\frac{1}{p} + \frac{1}{q} = 1$

def (convergenza debole). Siano  $f_n, f \in L^p(X)$ ,  $n \in \mathbb{N}$ ,  $1 \leq p < +\infty$ . Diciamo che  $f_n \xrightarrow[n \rightarrow \infty]{L^p\text{-debole}} f$  se  $\forall g \in L^q(X)$  si ha

$$\lim_{n \rightarrow \infty} \int_X f_n(x) g(x) d\mu = \int_X f(x) g(x) d\mu$$

oss Convergenza  $L^p$ -forte  $\Rightarrow$  convergenza  $L^p$ -debole

dim Sia  $f_n \xrightarrow[n \rightarrow \infty]{L^p\text{-forte}} f$ ,  $g \in L^q(X)$ . Allora

$$\left| \int_X (f_n(x) - f(x)) g(x) d\mu \right| \leq \int_X |f_n(x) - f(x)| \cdot |g(x)| d\mu$$

$$\stackrel{\text{Hölder}}{\leq} \|f_n - f\|_p \cdot \|g\|_q \xrightarrow[n \rightarrow \infty]{} 0. \quad \square$$

lemma Siano  $A \subseteq \mathbb{R}^n$  aperto con  $\mu = \mathcal{L}^n$  mis. di Lebesgue,  $(f_k)_{k \in \mathbb{N}}$  funzioni in  $L^p(A)$ ,  $1 < p < +\infty$  e  $\sup_{k \in \mathbb{N}} \|f_k\|_p < +\infty$ . Allora sono equivalenti:

A)  $f \xrightarrow[k \rightarrow \infty]{L^p\text{-debole}} f \in L^p(X)$

B)  $\lim_{k \rightarrow \infty} \int_A f_k(x) \varphi(x) dx = \int_A f(x) \varphi(x) dx \quad \forall \varphi \in C_c^\infty(A)$  (supporto compatto)

dim B)  $\Rightarrow$  A). Sia  $\varepsilon > 0$ ,  $g \in L^q(A)$ . Per il teorema di densità che vedremo più avanti  $\exists \varphi \in C_c^\infty(A)$

$$\text{e } \|g - \varphi\|_q < \varepsilon.$$

$$\left| \int_A f_k \cdot g dx - \int_A f \cdot g dx \right| \leq \int_A |f_k| |g - \varphi| dx + \left| \int_A (f_k - f) \varphi dx \right| + \int_A |f| |g - \varphi| dx$$

$$f_k g - f g = f_k (g - \varphi + \varphi) - f (g - \varphi + \varphi) = f_k (g - \varphi) + (f_k - f) \varphi + f (g - \varphi)$$

$$\stackrel{\text{Hölder}}{\leq} \|f_k\|_p \|g - \varphi\|_q + \left| \int_A (f_k - f) \varphi dx \right| + \|f\|_p \|g - \varphi\|_q$$

$\wedge \quad \wedge \quad \wedge$   
 $M < \infty \quad \varepsilon \quad \varepsilon$   
 $\wedge \quad \forall k \geq \bar{k} \exists \bar{k} \quad \wedge$

$$\leq (M + 1 + \|f\|_p) \varepsilon.$$

□

oss il teorema di densità non vale per  $q = +\infty$ , quindi nel lemma per  $p = 1$ .

oss convergenza debole  $\nrightarrow$  convergenza forte

esempio  $X = (0, \pi)$ ,  $1 < p < +\infty$ ,  $f_n(x) = \sin(nx)$ ,  $x \in (0, \pi)$ ,  $n \in \mathbb{N}$ .

•  $f_n \xrightarrow[n \rightarrow \infty]{L^p \text{ debole}} 0$ : sia  $\varphi \in C_c^\infty(0, \pi)$ , allora

$$\begin{aligned} \int_{(0, \pi)} f_n(x) \varphi(x) dx &= \int_0^\pi \sin(nx) \varphi(x) dx \\ &= \left[ -\frac{\cos(nx)}{n} \varphi(x) \right]_0^\pi + \int_0^\pi \frac{\cos(nx)}{n} \varphi'(x) dx \\ &= \frac{1}{n} \int_0^\pi \underbrace{\cos(nx)}_{-1 \leq \dots \leq 1} \underbrace{\varphi'(x)}_{\text{è quantità finita}} dx \xrightarrow[n \rightarrow \infty]{} 0 \end{aligned}$$

•  $f_n \not\xrightarrow[n \rightarrow \infty]{L^p \text{ forte}} 0$ :  $\int_0^\pi |f_n(x)|^p dx = \int_0^\pi |\sin(nx)|^p dx$

$$\stackrel{nx=t}{=} \int_0^{n\pi} \underbrace{|\sin t|^p}_{\text{periodica}} \frac{dt}{n}$$

$$= \int_0^\pi |\sin t|^p dt > 0$$

(non esistono neanche debolmente convergenti).

oss convergenza debole  $\nrightarrow$  convergenza puntuale

( $\nrightarrow$  esempio precedente)  
 $\nrightarrow$  prossimo esempio

esempio  $X = [0, 1]$ ,  $p = 1$ ,  $f_n(x) = n \chi_{(0, 1/n)}(x)$ ,  $x \in [0, 1]$ .

Osservo che  $f_n(x) \xrightarrow[n \rightarrow \infty]{} 0 \quad \forall x \in [0, 1]$  e  $\int_{(0, 1)} f_n(x) dx = 1 \quad \forall n \in \mathbb{N}^*$

$\forall g \in C([0, 1]) \supset C^\infty([0, 1])$

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_{(0, 1)} f_n(x) g(x) dx &= \lim_{n \rightarrow \infty} n \int_0^{1/n} g(x) dx = \lim_{n \rightarrow \infty} n (G(1/n) - G(0)) \\ &= \lim_{n \rightarrow \infty} \frac{G(1/n) - G(0)}{1/n} = g(0) \neq 0 \quad \text{in generale} \end{aligned}$$

Osserviamo infine che  $g(x) = \int_{[0,x]} g(x) d\delta_0$   
 $f_0 \mathcal{L}^1|_{[0,1]} \rightarrow f_0$  ↗ delta di Dirac centrato in 0

Esercizio Siano  $X = [0,1]$  con  $\mathcal{L}^1$  misura di Lebesgue,  $\varphi: [0,1] \rightarrow \mathbb{R}$  continua positiva (strettamente). Provare che  $\forall f \in \mathcal{L}^\infty([0,1])$  si ha che  $\lim_{p \rightarrow \infty} \left( \int_{[0,1]} |f(x)|^p \varphi(x) dx \right)^{1/p} = \|f\|_\infty$ .

• So che  $|f(x)| \leq \|f\|_\infty$  per q.o.  $x \in [0,1]$  (per def. di  $\|\cdot\|_\infty$ )

$$\Rightarrow \left( \int_{[0,1]} |f(x)|^p \varphi(x) dx \right)^{1/p} \leq \|f\|_\infty \underbrace{\left( \int_{[0,1]} \varphi(x) dx \right)^{1/p}}_{\in \mathbb{R}^+}$$

$$\Rightarrow \limsup_{p \rightarrow +\infty} \left( \int_{[0,1]} |f(x)|^p \varphi(x) dx \right)^{1/p} \leq \|f\|_\infty \cdot \lim_{p \rightarrow +\infty} \left( \int_{[0,1]} \varphi(x) dx \right)^{1/p} = \|f\|_\infty$$

• Per def.  $\|\cdot\|_\infty$ , fissa  $\varepsilon > 0$ :

$$\mathcal{L}^1(\underbrace{\{x \in [0,1] : |f(x)| \geq \|f\|_\infty - \varepsilon\}}_{:= A_\varepsilon}) > 0$$

$$\begin{aligned} \left( \int_{[0,1]} |f(x)|^p \varphi(x) dx \right)^{1/p} &\geq \left( \int_{A_\varepsilon} |f(x)|^p \varphi(x) dx \right)^{1/p} \geq (\|f\|_\infty - \varepsilon) \left( \int_{A_\varepsilon} \varphi(x) dx \right)^{1/p} \\ &\geq (\|f\|_\infty - \varepsilon) \left( \mathcal{L}^1(A_\varepsilon) \cdot \min_{x \in A_\varepsilon} \varphi(x) \right)^{1/p} \\ &\exists m = \min_{x \in [0,1]} \varphi(x) > 0 \end{aligned}$$

$$\Rightarrow \liminf_{p \rightarrow +\infty} \left( \int_{[0,1]} |f(x)|^p \varphi(x) dx \right)^{1/p} \geq (\|f\|_\infty - \varepsilon) \lim_{p \rightarrow +\infty} \left( \mathcal{L}^1(A_\varepsilon) \cdot m \right)^{1/p} = \|f\|_\infty - \varepsilon$$

$\forall \varepsilon > 0, \varepsilon < \|f\|_\infty$ .

•  $\Rightarrow$  per  $\varepsilon \rightarrow 0^+$ ,  $\|f\|_\infty \leq \liminf_{p \rightarrow +\infty} (\dots)^{1/p} \leq \limsup_{p \rightarrow +\infty} (\dots)^{1/p} \leq \|f\|_\infty$

$\Rightarrow \exists \lim_{p \rightarrow +\infty} (\dots)^{1/p} = \|f\|_\infty$ .

esercizio Provarne che  $\forall f \in C^1([0,1])$  con  $f(0) = f(1) = 0$ , vale  $\left(\int_0^1 f(x) dx\right)^2 \leq \frac{1}{12} \int_0^1 |f'(x)|^2 dx$   
 Inoltre, si determinino tutte le  $f$  per cui vale  $=$ .

$$\bullet \int_0^1 f(x) dx = \left[ \cancel{f(x)(x+\alpha)} \right]_0^1 - \int_0^1 f'(x)(x+\alpha) dx$$

$$= \int_0^1 (-x-\alpha) f'(x) dx$$

$$\stackrel{\text{Hölder}}{\leq} \left( \int_0^1 (x+\alpha)^2 dx \right)^{1/2} \left( \int_0^1 |f'(x)|^2 dx \right)^{1/2}$$

$$\Rightarrow \left( \int_0^1 f(x) dx \right)^2 \leq \int_0^1 |f'(x)|^2 dx \cdot \int_0^1 (x+\alpha)^2 dx, \quad \alpha \in \mathbb{R}$$

$$\bullet \int_0^1 (x+\alpha)^2 dx = \left[ \frac{(x+\alpha)^3}{3} \right]_0^1 = \frac{(1+\alpha)^3 - \alpha^3}{3} = \frac{1+3\alpha+3\alpha^2}{3} = (\alpha+1/2)^2 + 1/12 \stackrel{\alpha=-1/2}{\downarrow} = 1/12$$

$$\Rightarrow \left( \int_0^1 f(x) dx \right)^2 \leq 1/12 \int_0^1 |f'(x)|^2 dx$$

$\bullet L^1 = \text{vale} \Leftrightarrow L^p = \text{vale in Hölder}$

$$\Leftrightarrow \exists d \in \mathbb{R} \text{ t.c. } |g(x)| = |f(x)|^{p-1} \cdot d \text{ quando } \int f \cdot g \leq \|f\|^p \|g\|^q$$

$$\Leftrightarrow \exists d \in \mathbb{R} \text{ t.c. } f' = d(x-1/2)$$

$$\Leftrightarrow f(x) = \frac{d}{2} x^2 - \frac{d}{2} x + c \quad \exists d, c \in \mathbb{R}$$

Per ipotesi  $f(0) = 0 = f(1)$ , da cui  $f(x) = \frac{d}{2} x(x-1)$



# Convergenza in misura

def  $(X, \mathcal{A}, \mu)$ ,  $f, f_n \in \mathcal{M}(X)$ ,  $n \in \mathbb{N}$ . Diciamo che  $f_n \xrightarrow[n \rightarrow \infty]{\text{in misura}} f$  se:

$$\forall \varepsilon > 0 \quad \lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \varepsilon\}) = 0$$

(convergenza in misura).

## TEOREMA uellgono:

- 1) se  $\mu(X) < \infty$ , allora  $f_n(x) \xrightarrow[n \rightarrow \infty]{} f(x) \Rightarrow f_n \xrightarrow[n \rightarrow \infty]{\text{in misura}} f$  per  $\mu$ -q.o.  $x$
- 2) se  $f_n \xrightarrow[n \rightarrow \infty]{\text{in misura}} f \Rightarrow \exists$  sottoseq.  $(f_{n_k})_{k \in \mathbb{N}}$  t.c.  $f_{n_k}(x) \rightarrow f(x)$   $\mu$ -q.o.  $x \in X$

La dim di questo è nelle note (serve il lemma di Borel-Cantelli)

dim Prova 1). Sia  $\varepsilon > 0$  ed  $\eta > 0$ .

per Egoroff:  $\exists X_\eta \subset X$  t.c.  $\mu(X \setminus X_\eta) < \eta$  e  $f_n \xrightarrow[n \rightarrow \infty]{X_\eta} f$

Stime:

$$\mu(\{x \in X : |f_n(x) - f(x)| \geq \varepsilon\}) \leq \underbrace{\mu(X \setminus X_\eta)}_{\substack{\uparrow \\ \frac{1}{\varepsilon} \int_{X \setminus X_\eta} |f_n(x) - f(x)| d\mu \\ \uparrow \wedge \exists \bar{n} \text{ t.c. } \forall n \geq \bar{n} \\ \text{conv. unif.}}} + \mu(\{x \in X_\eta : |f_n(x) - f(x)| \geq \varepsilon\}) \leq 2\eta \quad \forall n \geq \bar{n}.$$

oss  $\overset{\text{L-forte}}{f_n \xrightarrow[n \rightarrow \infty]{} f} \not\Rightarrow f_n \xrightarrow[n \rightarrow \infty]{\text{in misura}} f$

Infatti:  $\mu(\{x \in X : |f_n(x) - f(x)| \geq \varepsilon\}) \leq \frac{1}{\varepsilon} \int_X |f_n(x) - f(x)| d\mu \xrightarrow{n \rightarrow \infty} 0$

Per la direzione opposta: considero  $X = (0, 1)$ ,  $f_n(x) = nX_{(0, 1/n)}(x)$ , e osservo che non è nemmeno possibile avere una successione che converge in  $L^1$ .

## TEOREMA $\mu(X) < \infty$ , $f, f_n \in \mathcal{M}(X)$ , $n \in \mathbb{N}$ . Sono equivalenti:

- A)  $f_n \xrightarrow[n \rightarrow \infty]{\text{in misura}} f$
- B)  $\lim_{n \rightarrow \infty} \int_X \frac{|f_n(x) - f(x)|}{1 + |f_n(x) - f(x)|} d\mu = 0$

wlog

dim A)  $\Rightarrow$  B)  $f=0$ .  $\int_X \frac{|f_n(x)|}{1 + |f_n(x)|} d\mu \stackrel{\substack{\text{integratore per sopralivelli} \\ (\text{vale per ogni m.f.: Coroll. FT})}}{=} \int_0^1 \mu(\{x \in X : \frac{|f_n(x)|}{1 + |f_n(x)|} \geq t\}) dt$

$\Leftrightarrow |f_n(x)|(1-t) \geq t$

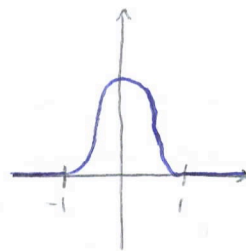
$\lim_{n \rightarrow \infty} \int_X \frac{|f_n(x)|}{1 + |f_n(x)|} d\mu \stackrel{\substack{\text{cap. dominante} \\ \text{poich\'e l'integranda \'} \\ < \mu(X) < \infty}}{=} \lim_{n \rightarrow \infty} \int_0^1 \underbrace{\mu(\{x \in X : |f_n(x)| \geq \frac{t}{1-t}\})}_{= f_n(t) \xrightarrow[n \rightarrow \infty]{} 0 \text{ dalla conv. in misura}} dt = \int_0^1 \lim_{n \rightarrow \infty} (\dots) dt = 0$

seguo B)  $\Rightarrow$  A).

## Regularization

$X = \mathbb{R}^n$ ,  $\mu = \mathcal{L}^n$ . Sia  $\chi: \mathbb{R}^n \rightarrow [0, +\infty)$

$$\chi(x) = \begin{cases} C_0 \exp\left(\frac{1}{|x|^2 - 1}\right) & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$



ora regolarizziamo il  
adattare la palla unitaria e  
e far tendere la 0 all'infinito  
(approssimando la delta di Dirac).

Osserviamo che  $\chi \in C^\infty(\mathbb{R}^n)$ . Fatto  $C_0 > 0$  e  $\int_{\mathbb{R}^n} \chi(x) dx = 1$ .

Per  $\varepsilon > 0$  sia  $\chi_\varepsilon(x) = \frac{1}{\varepsilon^n} \chi\left(\frac{x}{\varepsilon}\right)$ ,  $x \in \mathbb{R}^n$

Proprietà:

1)  $\chi_\varepsilon \in C^\infty(\mathbb{R}^n)$ , radiali

2)  $\chi_\varepsilon \geq 0$  e  $\chi_\varepsilon(x) > 0 \iff |x| < \varepsilon$

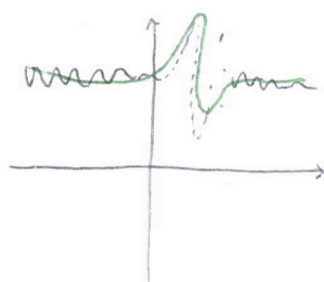
3) Inoltre  $\int_{\mathbb{R}^n} \frac{1}{\varepsilon^n} \chi\left(\frac{x}{\varepsilon}\right) dx = 1$ , infatti:  
 $x = \varepsilon y$ ,  $dx = \varepsilon^n dy$   
 $\int_{\mathbb{R}^n} \frac{1}{\varepsilon^n} \chi\left(\frac{x}{\varepsilon}\right) dx = \int_{\mathbb{R}^n} \chi(y) dy = 1$

Sia ora  $f \in L^1_{loc}(\mathbb{R}^n)$ , ovvero  $f \in L^1(K)$  per ogni  $K \subset \mathbb{R}^n$  compatto.

Def. Siano  $\varepsilon > 0$  ed  $f \in L^1_{loc}(\mathbb{R}^n)$ . Definiamo la **REGOLARIZZAZIONE** di  $f$  come:

$$f_\varepsilon(x) = \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) f(y) dy =: \chi_\varepsilon * f(x)$$

$\nwarrow$  **PRODOTTO DI CONVOLUZIONE**



$f$  irregolare, brutta  
con la convoluzione la rendiamo liscia

oss  $f_\varepsilon(x) = \chi_\varepsilon * f(x) = \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) f(y) dy$   
 $x-y=y'$   
 $= \int_{\mathbb{R}^n} \chi_\varepsilon(y) f(x-y) dy = f * \chi_\varepsilon(x)$

notazione  $\text{pt}(f) = \overline{\{x \in \mathbb{R}^n : f(x) \neq 0\}}$

**TEOREMA** Sia  $f \in L^1_{loc}(\mathbb{R}^n)$ . Allora:

i)  $f_\varepsilon \in C^\infty(\mathbb{R}^n)$ ,  $\forall \varepsilon > 0$

ii) Detto  $K = \text{pt}(f)$  avremo  $\text{pt}(f_\varepsilon) \subset K_\varepsilon := \{x \in \mathbb{R}^n : \text{dist}(x, K) \leq \varepsilon\}$

iii) se  $f \in C(\mathbb{R}^n)$ , allora  $f_\varepsilon \xrightarrow[\varepsilon \rightarrow 0^+]{K} f$  con  $K \subset \mathbb{R}^n$  compatto

iv) se poi  $f \in L^p(\mathbb{R}^n)$ ,  $1 \leq p \leq \infty$ , allora  $\|f_\varepsilon\|_p \leq \|f\|_p$ .

dim i) Affermo che  $f_\varepsilon$  è continua su  $\mathbb{R}^n$ , ovvero in  $x_0 \in \mathbb{R}^n$  generico.

Premessa:  $|x - x_0| \leq 1$ . Allora

$$\text{supp } \chi_\varepsilon(x - \varepsilon) \subset \overline{B}(x_0) = K \text{ compatto}$$

Allora

$$\lim_{x \rightarrow x_0} f_\varepsilon(x) = \lim_{x \rightarrow x_0} \int_{\mathbb{R}^n} \underbrace{\chi_\varepsilon(x-y)}_{\in L^\infty} \underbrace{f(y)}_{\in L^1(K)} dy \Rightarrow \|\chi_\varepsilon\|_\infty \|f\|_1 \in L^1(K)$$

porro  
dentro a  
unite  $\searrow$

$$= \int_{\mathbb{R}^n} \chi_\varepsilon(x_0 - y) f(y) dy$$

$$= f_\varepsilon(x_0) \quad \text{particolare}$$

$i=1, \dots, n$ , mostro che le derivate  $\partial_{x_i} f_\varepsilon$  erano continue.

$$\frac{\partial}{\partial x_i} f_\varepsilon(x) = \frac{\partial}{\partial x_i} \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) f(y) dy \stackrel{\text{coll. conv. dominata}}{=} \int_{\mathbb{R}^n} \left( \frac{\partial}{\partial x_i} \chi_\varepsilon \right)(x-y) f(y) dy \text{ è continua,}$$

con ragionamento analogo a quello sopra. E iterando si trova  $f \in C^\infty(\mathbb{R}^n)$ .

ii) se  $x \notin K_\varepsilon$  avremo  $\text{dist}(x, K) > \varepsilon$  con  $K = \text{supp}(f)$

$$f_\varepsilon(x) = \int_{\substack{K \\ \cup \\ \emptyset}} \chi_\varepsilon(x-y) f(y) dy = 0$$

iii)  $K \subset \mathbb{R}^n$  compatto. scelgo  $\varepsilon \leq 1$

$$\begin{aligned} f_\varepsilon(x) - f(x) &= \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) f(y) dy - f(x) \overbrace{\int_{\mathbb{R}^n} \chi_\varepsilon(x-y) dy}^{=1} \\ &= \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) (f(y) - f(x)) dy \end{aligned}$$

$K_1 = \{x \in \mathbb{R}^n : \text{dist}(x, K) \leq 1\}$  compatto

$f$  è unif. cont. su  $K_1$ , ovvero  $\forall \sigma > 0 \exists \varepsilon_0 > 0$  t.c.  $|x-y| < \varepsilon_0 \Rightarrow |f(x) - f(y)| < \sigma$

$x \in K$ ,  $\varepsilon \leq 1$ ,  $y \in K_1$

$$|f_\varepsilon(x) - f(x)| \leq \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) \overset{\sigma \text{ poiché } y \in K_1}{|f(y) - f(x)|} dy < \sigma \quad \text{per } \varepsilon < \varepsilon_0.$$

iv) caso  $1 < p < +\infty$ .

$$|f_\varepsilon(x)| \leq \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) |f(y)| dy$$

$$= \int_{\mathbb{R}^n} \underbrace{\chi_\varepsilon(x-y)^{1/q}}_{\in L^q} \underbrace{\chi_\varepsilon(x-y)^{1/p}}_{\in L^p} |f(y)| dy$$

Hölder

$$\leq \left( \underbrace{\int_{\mathbb{R}^n} \chi_\varepsilon(x-y) dy}_{=1} \right)^{1/q} \cdot \left( \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) |f(y)|^p dy \right)^{1/p}$$

$$\Rightarrow |f_\varepsilon(x)|^p \leq \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) |f(y)|^p dy \quad \forall x \in \mathbb{R}^n$$

$$\Rightarrow \int_{\mathbb{R}^n} |f_\varepsilon(x)|^p dx \leq \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^n} \chi(x-y) |f(y)|^p dy \right) dx$$

$$\stackrel{FT}{=} \int_{\mathbb{R}^n} |f(y)|^p \underbrace{\left( \int_{\mathbb{R}^n} \chi(x-y) dx \right)}_{=1} dy$$

$$= \int_{\mathbb{R}^n} |f(y)|^p dy = \|f\|_p^p$$

$$\Rightarrow \|f_\varepsilon\|_p \leq \|f\|_p$$

□

obiettivo  $f_\varepsilon \xrightarrow[\varepsilon \rightarrow 0^+]{L^p} f$

### Teorema di densità

Sia  $A \subseteq \mathbb{R}^n$  aperto,  $C_c(A) = \{ f \in C(A) : \text{spt}(f) \overset{A}{\subset} \text{compatto} \}$

TEOREMA  $\forall p \in [1, \infty)$ ,  $\overline{C_c(A)}^{L^p(A)} = L^p(A)$  con  $A \subseteq \mathbb{R}^n$  aperto.

dim Dimostriamo che data  $f \in L^p(A)$   $\forall \varepsilon > 0 \exists g \in C_c(A)$  t.c.  $\|f - g\|_p < \varepsilon$ .

wlog  $\text{spt}(f) \subseteq A$  compatto ( $f_n(x) = f(x) \cdot \chi_{K_n}(x)$ ,  $K_n = \{x \in A : |x| \leq n \text{ e } \text{dist}(x, \partial A) \geq \frac{1}{n}\}$ )

e poi si usa il teorema di conv. dominata su  $\|f - g\|_p \leq \|f - f_k\|_p + \|f_k - g\|_p$ .

wlog  $f \geq 0$  ( $f = f^+ - f^-$ ).

wlog  $f$  semplice con  $\text{spt}(f) \subseteq A$  compatto (dualità di approx.  $\exists \varphi$  semplice t.c.  $\int_A |f(x) - \varphi(x)|^p dx < \varepsilon$ ).

$\Rightarrow$  sia  $f(x) = \sum_{k=1, \dots, K} c_k \chi_{A_k}(x)$ ,  $A_k \subseteq \mathbb{R}^n$  misurabile disgiunti,  $c_k \in \mathbb{R} \geq 0$ .

wlog  $f = \chi_E$ ,  $E$  misurabile con  $\bar{E}$  compatto  $\subseteq A$

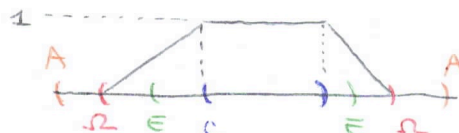
Per regolarità di  $\mathcal{L}^n$ , dato  $\varepsilon > 0 \exists C \subseteq E \subseteq \Omega$  t.c.  $\mathcal{L}^n(\Omega \setminus C) < \varepsilon$ .

Ona definisco  $g: A \rightarrow [0, +\infty]$ ,  $g(x) = \frac{\text{dist}(x; \mathbb{R}^n \setminus \Omega)}{\text{dist}(x; \mathbb{R}^n \setminus \Omega) + \text{dist}(x; C)}$

1)  $g$  è ben definita:  $\text{dist}(x; \mathbb{R}^n \setminus \Omega) \geq 0$ , ed  $e = 0 \Leftrightarrow x \in \mathbb{R}^n \setminus \Omega$   
 $\text{dist}(x; C) \geq 0$ , ed  $e = 0 \Leftrightarrow x \in C$

2)  $g \in C_c(A)$

3)  $0 \leq g \leq 1$ ,  $g \equiv 1$  su  $C$ ,  $g \equiv 0$  su  $\mathbb{R}^n \setminus \Omega$



$$\Rightarrow \int_A |\chi_E(x) - g(x)|^p dx = \int_C \underbrace{|\chi_E - \hat{1}|^p}_{=0} dx + \int_{\Omega \setminus C} \underbrace{|\chi_E - g|}_{\hat{1}}^p dx + \int_{A \setminus \Omega} \underbrace{|\chi_E - g|}_{=0}^p dx$$

$$\leq \mathcal{L}^n(\Omega \setminus C) \leq \varepsilon.$$



# Convergenza in media e corollari

**TEOREMA** Siano  $1 \leq p < \infty$ ,  $f \in L^p(\mathbb{R}^n)$ , allora sono vere:

- 1)  $\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} |f(x+h) - f(x)|^p dx = 0$  (continuità in media)
- 2)  $f \xrightarrow[\varepsilon \rightarrow 0^+]{L^p\text{-forte}} f$

dim 1) Sia  $\sigma > 0$ :  $\exists g \in C_c(\mathbb{R}^n)$  t.c.  $\|f - g\|_p < \sigma$ .

Sia  $K = \text{supp}(g)$  e  $K_1 = \{x \in \mathbb{R}^n : \text{dist}(x, K) \leq 1\}$  è compatto.

$g$  è unif. continua su  $\mathbb{R}^n \Rightarrow \exists \varepsilon > 0$  t.c.  $\forall x \in K$  e  $|h| < \varepsilon \leq 1$ ,  $|g(x+h) - g(x)| < \sigma$

Adesso stimiamo:

$$\begin{aligned} \left( \int_{\mathbb{R}^n} |f(x+h) - f(x)|^p dx \right)^{1/p} &\leq \underbrace{\left( \int_{\mathbb{R}^n} |f(x+h) - g(x+h)|^p dx \right)^{1/p}}_{< \sigma} + \underbrace{\left( \int_{\mathbb{R}^n} |g(x+h) - g(x)|^p dx \right)^{1/p}}_{< \sigma} + \underbrace{\left( \int_{\mathbb{R}^n} |g(x) - f(x)|^p dx \right)^{1/p}}_{< \sigma} \\ &= \underbrace{\left( \int_{K_1} |g(x+h) - g(x)|^p dx \right)^{1/p}}_{< \sigma \text{ per } |h| < \varepsilon \leq 1} + \underbrace{\left( \int_{\mathbb{R}^n} |g(x) - f(x)|^p dx \right)^{1/p}}_{< \sigma} \\ &\leq 2\sigma + \mathcal{L}^n(K_1)^{1/p} \sigma = \sigma (2 + \mathcal{L}^n(K_1)^{1/p}) = C \cdot \sigma, C \in \mathbb{R} \end{aligned}$$

$$\begin{aligned} 2) f_\varepsilon(x) &= \int_{\mathbb{R}^n} \chi_\varepsilon(x-y) f(y) dy = \int_{\mathbb{R}^n} f(x-y) \chi_\varepsilon(y) dy \\ &= \int_{\mathbb{R}^n} \chi_\varepsilon(x+h) \underbrace{\chi_\varepsilon(-h)}_{=\chi_\varepsilon(h)} dh = \int_{B_\varepsilon(0)} f(x+h) \chi_\varepsilon(h) dh \end{aligned}$$

$$f(x) = \int_{\mathbb{R}^n} f(x) \chi_\varepsilon(h) dh$$

$$\begin{aligned} |f_\varepsilon(x) - f(x)| &= \left| \int_{B_\varepsilon(0)} \underbrace{\chi_\varepsilon(h)}_{\substack{L^p \\ \chi_\varepsilon(h)^{1/p} \cdot \chi_\varepsilon(h)^{1/q} \in L^q}} (f(x+h) - f(x)) dh \right| \\ &\stackrel{\text{Hölder}}{\leq} \underbrace{\left( \int_{B_\varepsilon(0)} \chi_\varepsilon(h) dh \right)^{1/q}}_{=1} \left( \int_{B_\varepsilon(0)} |f(x+h) - f(x)|^p \chi_\varepsilon(h) dh \right)^{1/p} \end{aligned}$$

$$\begin{aligned} \|f_\varepsilon - f\|_p^p &= \int_{\mathbb{R}^n} |f_\varepsilon(x) - f(x)|^p dx \\ &\leq \int_{\mathbb{R}^n} \left( \int_{B_\varepsilon(0)} |f(x+h) - f(x)|^p \chi_\varepsilon(h) dh \right) dx \\ &\stackrel{\text{FT}}{=} \int_{B_\varepsilon(0)} \chi_\varepsilon(h) \left( \int_{\mathbb{R}^n} |f(x+h) - f(x)|^p dx \right) dh \\ &\leq \sigma^p \quad \forall |h| \leq \varepsilon \text{ per 1) (continuità in media).} \end{aligned}$$

$$\Rightarrow \forall \sigma > 0 \exists \varepsilon_0 > 0 \text{ t.c. } \forall \varepsilon \in (0, \varepsilon_0) \quad \|f_\varepsilon - f\|_p < \sigma. \quad \square$$

corollario  $\forall p \in [1, +\infty)$ ,  $\overline{C_c^\infty(A)}^{L^p(A)} = L^p(A)$  (con  $A \subseteq \mathbb{R}^n$  aperto).

dim (idea)  $\overline{C_c(A)} = L^p(A)$ ;  $\forall f \in C_c(A)$ , le regolanti.  $f_\varepsilon \in C_c^\infty(A)$  convergono in  $L^p$ -forte



TEOREMA (Riemann-Lebesgue).  $f \in L^1(\mathbb{R}^n)$ ,  $\hat{f}: \mathbb{R}^n \rightarrow \mathbb{C}$ ,  $\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) \cdot e^{2\pi i \xi \cdot x} dx$   
 $(\hat{f}(\xi))$  converge ed è continua per C.D).  $\forall f \in L^1(\mathbb{R}^n)$ ,  $\lim_{|\xi| \rightarrow +\infty} \hat{f}(\xi) = 0$ .

dim ( $n=1$ ).

$$\forall \varepsilon > 0 \exists g \in C_c^\infty(\mathbb{R}) \text{ t.c. } \int_{\mathbb{R}} |f(x) - g(x)| dx < \varepsilon$$

$$|\hat{f}(\xi)| \leq |\hat{f}(\xi) - \hat{g}(\xi)| + |\hat{g}(\xi)|$$

$$= \underbrace{\left| \int_{\mathbb{R}} (f(x) - g(x)) e^{2\pi i \xi x} dx \right|}_m + \underbrace{\left| \int_{\mathbb{R}} g(x) e^{2\pi i \xi x} dx \right|}_n$$

$$\int_{\mathbb{R}} |f(x) - g(x)| dx = \|f - g\|_1 \quad \left| \left[ \frac{g(x) e^{2\pi i \xi x}}{2\pi i \xi} \right]_{-\infty}^{+\infty} - \int_{\mathbb{R}} \frac{g'(x) e^{2\pi i \xi x}}{2\pi i \xi} dx \right|$$

$$\int_{\mathbb{R}} |g'(x)| \frac{|e^{2\pi i \xi x}|}{|2\pi i \xi|} dx = \frac{1}{2\pi |\xi|} \int_{\mathbb{R}} |g'(x)| dx$$

$$\leq \|f - g\|_1 + \frac{1}{2\pi |\xi|} \int_{\mathbb{R}} |g'(x)| dx \xrightarrow{|\xi| \rightarrow \pm\infty} \|f - g\|_1 \quad \square$$

esercizio  $X = [0,1]$ ,  $\mu = \mathcal{L}^1([0,1])$ ,  $1 \leq p < +\infty$ ,  $\varepsilon_n \in [0,1]$  misurabili;  
 $f = \chi_\varepsilon$  ed  $f_n = \chi_{\varepsilon_n}$ ,  $n \in \mathbb{N}$ . Provare che  $f_n \xrightarrow[n \rightarrow \infty]{\mathcal{L}\text{-forte}} f \iff f_n \xrightarrow[n \rightarrow \infty]{\mathcal{L}\text{-debole}} f$

( $\implies$ ) sempre vera.

( $\impliedby$ ) Suppongo  $f_n \xrightarrow[n \rightarrow \infty]{\mathcal{L}\text{-debole}} f$ , cioè  $\forall g \in \mathcal{L}^q([0,1])$  con  $\frac{1}{p} + \frac{1}{q} = 1$

$$\lim_{n \rightarrow \infty} \int_{[0,1]} f_n g dx = \int_{[0,1]} f g dx$$

$$\int_{[0,1]} \underbrace{|f_n - f|}_p dx = \int_{[0,1]} |f_n - f|^2 dx$$

$$f_n - f = \begin{cases} 1 \\ 0 \\ -1 \end{cases} = \int_{[0,1]} (f_n^2 - 2f_n f + f^2) dx$$

$$= \int_{[0,1]} f_n (1 - 2f) dx + \int_{[0,1]} f dx$$

$$\xrightarrow{n \rightarrow \infty} \int_{[0,1]} f (1 - 2f) dx + \int_{[0,1]} f dx$$

$$= \int_{[0,1]} (-f) dx + \int_{[0,1]} f dx = 0$$

esercizio  $X = \mathbb{R}^+$ ,  $\mu = \mathcal{L}^1|_{\mathbb{R}^+}$ ,  $f_n(x) = \sqrt{n} \chi_{(e_n(n), e_n(n+1))}(x)$ ,  $n \in \mathbb{N}^+$ .

1) Per quali  $p$ ,  $1 \leq p \leq \infty$ ,  $f_n$  converge fortemente in  $L^p(\mathbb{R}^+)$ ?

Osservo che  $f_n(x) \xrightarrow{n \rightarrow \infty} 0 \ \forall x$ , dunque se  $f_n$  converge, deve convergere a 0 per  $\mu$ -q.o.  $x \in \mathbb{R}$

$$\|f_n - 0\|_p^p = \int_{\mathbb{R}^+} f_n^p dx = \int_{e_n(n)}^{e_n(n+1)} n^{p/2} dx = n^{p/2} e_n\left(\frac{n+1}{n}\right) = n^{p/2} e_n\left(1 + \frac{1}{n}\right)$$

$$\sim n^{p/2} \cdot \frac{1}{n} = n^{p/2-1} \xrightarrow{n \rightarrow +\infty} 0$$

$p < 2 \iff$

2) Per quali  $p$ ,  $1 \leq p \leq +\infty$ ,  $f_n$  converge debolmente in  $L^p(\mathbb{R}^+)$ ?

Ricordo conv. forte  $\Rightarrow$  conv. debole  $\Rightarrow f_n \xrightarrow[n \rightarrow +\infty]{L^p\text{-deb}} 0 \ \forall p < 2$ .

Se  $f_n \xrightarrow{L^p\text{-deb}} f$ ,  $\int_{\mathbb{R}^+} f_n g dx \rightarrow \int_{\mathbb{R}^+} f g dx \ \forall g \in \mathcal{L}^q(\mathbb{R}^+)$ .

• Fisso  $M > 0$  "grande", scelgo  $g \in \mathcal{L}^q(\mathbb{R}^+)$  con  $\text{supp}(g) \subset [0, M]$ ,

$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^+} f_n g dx = 0 \Rightarrow$  deve essere  $f(x) = 0$  per  $\mu$ -q.o.  $x \in X$  in  $[0, M]$

$\forall M > 0 \Rightarrow f = 0$  per  $\mu$ -q.o.  $x \in X$ . Dunque  $f_n \xrightarrow[n \rightarrow \infty]{L^p\text{-deb}} f$ ,  $f \equiv 0$ .

• Per quali  $q$  si ha  $\int_{\mathbb{R}^+} f_n g dx \xrightarrow{n \rightarrow \infty} 0 \ \forall g \in \mathcal{L}^q(\mathbb{R}^+)$

ha  $\varepsilon > 0$ ,  $g \in \mathcal{L}^q(\mathbb{R}^+)$ ,  $M > 0$ ,  $\int_M^{+\infty} |g(x)|^q dx < +\infty$ ,  $\lim_{m \rightarrow \infty} \int_m^{+\infty} |g(x)|^q dx = 0$

$\Rightarrow \int_m^{+\infty} |g(x)|^p dx < \varepsilon^q \ \forall M > M_0, \exists M_0$ .

$$\Rightarrow 0 \leq \int_{\mathbb{R}^+} f_n(x) |g(x)| dx \stackrel{\text{Hölder}}{\leq} \left( \int_{\mathbb{R}^+} f_n^p dx \right)^{1/p} \left( \int_M^{+\infty} |g(x)|^q dx \right)^{1/q} \leq C \cdot \varepsilon$$

$\uparrow$   
per  $n > M_0$        $\uparrow$   
 $\infty \ \forall p \leq 2$

$\Rightarrow f_n \xrightarrow{L^p\text{-debole}} 0 \ \forall p \leq 2$ .

esercizio caso  $p > 2$ .

## Lemma su separabilità

$(X, d)$  si dice **SEPARABILE** se  $\exists x_0 \subset X$  tale che  $\text{card}(X_0) = \text{card}(\mathbb{N})$  e  $\overline{X_0} = X$

**TEOREMA** Siano  $A \subset \mathbb{R}^n$  aperto e  $1 \leq p < \infty$ . Allora  $L^p(A)$  è separabile.

**TEOREMA**  $L^\infty(A)$  non è separabile.

dim  $A = \mathbb{R}^n$ ,  $r > 0$ ,  $B_r = \{x \in \mathbb{R}^n : |x| \leq r\}$

$$X_{B_r} \in L^\infty(A), \quad B_r = \{f \in L^\infty(A) : \|f - X_{B_r}\|_\infty \leq 1/2\}$$

Osservo:  $r \neq s$ ,  $B_r \cap B_s = \emptyset$

Sia  $\mathcal{F} \subset L^\infty(\mathbb{R}^n)$  denso,  $\overline{\mathcal{F}} = L^\infty(\mathbb{R}^n)$ .

Ora  $\forall r > 0 \exists f_r \in B_r$  per densità.

Sia poi  $\Phi: \mathbb{R}^+ \rightarrow \mathcal{F}$ ,  $\Phi(r) = f_r$ : è funzione iniettiva

$\Rightarrow \text{card}(\mathbb{R}^+) \leq \text{card}(\mathcal{F}) \Rightarrow \mathcal{F}$  non è numerabile.  $\square$

$$\text{PA } \emptyset \neq B_r \cap B_s \exists f$$
$$1 = \|X_{B_r} - X_{B_s}\|_\infty \leq \underbrace{\|X_{B_r} - f\|_\infty}_{1/2} + \underbrace{\|X_{B_s} - f\|_\infty}_{1/2} < 1 \quad \text{↯}$$

## Dualità

Sia  $(X, \|\cdot\|)$  SN.

def Il **DUALE** di  $X$  è  $X^* = \{T: X \rightarrow \mathbb{R} \text{ lineare e continua/limitata}\}$

$\overset{\cup}{T}$  ha norma  $\|T\| = \sup_{\|x\| \leq 1} |Tx| < \infty$ , detta **NORMA OPERATORIA**

in questo caso sono  
la stessa cosa

$(X, \mathcal{A}, \mu)$ ,  $1 \leq p < \infty$  e  $1 < q \leq \infty$ ,  $\frac{1}{p} + \frac{1}{q} = 1$

Fisso  $g \in L^q(X)$ . Definisco  $T_g: L^p(X) \rightarrow \mathbb{R}$

$$T_g(f) = \int_X f(x)g(x) d\mu$$

essa è lineare ed è anche unitaria:

$$|T_g(f)| \leq \|g\|_q \cdot \|f\|_p \Rightarrow \|T_g\| \leq \|g\|_q < \infty$$

$\nwarrow$  Hölder

$\Rightarrow g \in L^p(X)^*$

Affermo che  $\|T_g\| = \|g\|_q$ .

Scelgo  $f(x) = \frac{|g(x)|^{\frac{q}{p}}}{\|g\|_q^{\frac{q}{p}}} \cdot \text{sgn}(g(x))$ ,  $g \neq 0$ :

Ora

$$\int_X |f(x)|^p d\mu = \frac{1}{\|g\|_q^q} \int_X |g(x)|^q d\mu = 1$$

Conti:

$$T_g(f) = \frac{1}{\|g\|_q^{\frac{q}{p}}} \int_X |g(x)|^{\frac{q}{p}+1} d\mu = \frac{1}{\|g\|_q^{\frac{q}{p}}} \int_X |g(x)|^q d\mu = \|g\|_q^{\frac{q}{p}+1} = \|g\|_q^q = \|g\|_q^q$$

$\frac{q}{p}+1 = q(\frac{1}{p}+\frac{1}{q}) = q$        $\frac{q}{p} = q(1-\frac{1}{q}) = q \cdot \frac{1}{q} = 1$

**TEOREMA**  $(X, \mathcal{A}, \mu)$  spazio di misura,  $1 \leq p < \infty$  e  $\frac{1}{p} + \frac{1}{q} = 1$ ,  $q \in [1, \infty]$ . Allora

$$L^p(X)^* = L^q(X)$$

con identificazione lineare ed isometrica.

Per  $p=1$  si assume che  $\mu$  sia  $\sigma$ -finita.

dim Per  $T \in L^p(X)^*$  voglio trovare  $h \in L^q(X)$  con

$$T(f) = \int_X f(x) h(x) d\mu \quad \text{e} \quad \|T\| = \|h\|_q$$

Dimostrazione nel caso:

$$1) \mu(X) < \infty$$

$$2) f \geq 0 \Rightarrow T(f) \geq 0$$

$$\text{Sia } d: \mathcal{A} \rightarrow [0, \infty), \quad d(A) = T(\chi_A) \geq 0$$

Affermo che  $d$  è  $\sigma$ -additiva.

$$\text{Sia } A = \bigcup_{n=1}^{\infty} A_n \text{ disgiunta con } A_n \in \mathcal{A}. \quad \chi_A = \sum_{n=1}^{\infty} \chi_{A_n}$$

$$\text{Ora } d(A) = T(\chi_A) = T\left(\sum_{n=1}^{\infty} \chi_{A_n}\right) = T\left(\lim_{n \rightarrow \infty} \sum_{k=1}^n \chi_{A_k}\right)$$

La differenza è  $\leq 2$   
e poi vale C.D

Osservo che  $\sum_{k=1}^n \chi_{A_k}(x) \xrightarrow[n \rightarrow \infty]{L^p\text{-forte}} \chi_A(x)$  e  $T$  continua, dunque

$$d(A) = \dots = \lim_{n \rightarrow \infty} T\left(\sum_{k=1}^n \chi_{A_k}\right) \underset{T \text{ lineare}}{=} \lim_{n \rightarrow \infty} \sum_{k=1}^n T(\chi_{A_k}) = \sum_{k=1}^{\infty} d(A_k)$$

Affermo  $d \ll \mu$ , infatti per  $A \in \mathcal{A}$ : def. di norma operatoriale

$$0 \leq d(A) = T(\chi_A) \leq \|T\| \cdot \|\chi_A\|_p = \|T\| (\mu(A))^{1/p}$$

Per il teorema di Radon-Nykodym (che proveremo) esiste  $h: X \rightarrow [0, \infty)$  misurabile tale che

$$d(A) = \int_A h(x) d\mu.$$

$$\bullet \varphi = \sum_{k=1}^n c_k \chi_{A_k}, \quad A_k \in \mathcal{A}$$

$$\begin{aligned} T(\varphi) &= T\left(\sum_{k=1}^n c_k \chi_{A_k}\right) = \sum_{k=1}^n c_k T(\chi_{A_k}) \\ &= \sum_{k=1}^n c_k d(A_k) = \sum_{k=1}^n c_k \int_X \chi_{A_k} h d\mu \\ &= \int_X \varphi(x) h(x) d\mu \end{aligned}$$

$$\bullet f \in L^p(X), \text{ wlog } f \geq 0$$

do che  $\exists \varphi_n$  funzioni semplici tali che  $0 \leq \varphi_n \leq \varphi_{n+1} \uparrow f$  su  $X$

$$\Rightarrow \varphi_n \xrightarrow[n \rightarrow \infty]{L^p\text{-forte}} f \quad \text{convergenza } L^p\text{-forte (in norma) e } T \text{ continua}$$

$$T(f) = \lim_{n \rightarrow \infty} T(\varphi_n) = \lim_{n \rightarrow \infty} \int_X \varphi_n(x) h(x) d\mu$$

$$\stackrel{\text{C.M.}}{=} \int_X f(x) h(x) d\mu$$

$$\|T\| = \sup_{\|f\|_p \leq 1} |T(f)| \stackrel{\text{Hölder}}{\leq} \|h\|_{L^q(X)}$$

Voglio provare  $\|h\|_{L^q(X)} \leq \|T\|$ .

$$1) \text{ caso } 1 < p < \infty.$$

Esiste successione di funzioni semplici  $0 \leq \varphi_n \leq \varphi_{n+1} \uparrow h$  su  $X$

$$\text{Sia } k > 0 \text{ da scegliere} \quad T(\varphi_n^k) = \int_X \varphi_n^k(x) h(x) d\mu \geq \int_X \varphi_n(x)^{k+1} d\mu$$

le  $\varphi_n$  stanno in tutti gli  $L^p$  ed  $L^q$



$$T(\varphi_n^k) \leq \|T\| \cdot \|\varphi_n^k\|_p = \|T\| \cdot \left( \int_X \varphi_n^{kp}(x) d\mu \right)^{\frac{1}{p}}$$

← def. di norma operatoriale

Scelgo  $kp = k+1 \Leftrightarrow k = \frac{1}{p-1}$

$$\Rightarrow \|T\| \geq \left( \int_X \varphi_n(x) \left( \frac{1}{p-1} + 1 \right) d\mu \right)^{\frac{p}{p-1}} \left( 1 - \frac{1}{p} \right) = \frac{p-1}{p}, \text{ ma } \frac{p}{p-1} = q$$

$$\Rightarrow \|T\| \geq \left( \int_X \varphi_n(x)^q d\mu \right)^{\frac{1}{q}}$$

$$\stackrel{cm}{\Rightarrow} \text{ per } n \rightarrow +\infty \quad \|T\| \geq \|h\|_q$$

$$\Rightarrow \|T\| = \|h\|_q$$

$$\|T\| \left( \int_X \varphi_n^{kp}(x) d\mu \right)^{\frac{1}{p}} \geq \int_X \varphi_n^{k+1} d\mu$$

$$\|T\| \left( \int_X \varphi_n^{\frac{p}{p-1}} d\mu \right)^{\frac{1}{p}} \geq \int_X \varphi_n^{\frac{p}{p-1}} d\mu$$

2) caso  $p=1$ , o  $q=\infty$ .

Bisogna provare che

$$\|T\| \geq \|h\|_\infty$$

Stimiamo

$$\mu \left( \underbrace{\left\{ x \in X : |h(x)| \geq \|T\| + \frac{1}{n} \right\}}_{= A_n} \right) \stackrel{?}{=} 0$$

$$T(\chi_{A_n}) = \int_{A_n} h(x) d\mu \geq \left( \|T\| + \frac{1}{n} \right) \mu(A_n)$$

$$T(\chi_{A_n}) \leq \|T\| \|\chi_{A_n}\|_1 = \|T\| \mu(A_n)$$

$$\Rightarrow \|T\| \mu(A_n) \leq \left( \|T\| + \frac{1}{n} \right) \mu(A_n) \Leftrightarrow \mu(A_n) = 0 \quad \forall n > 0. \quad \square$$